

15 Oct

15 October 2025 10:08

PCA

population:  $\Sigma = P \Lambda P^T$  ( $\lambda_1 > \lambda_2 > \dots > \lambda_p > 0$ )

$$y_j = a_j^T X \quad 1 \leq j \leq p \quad \text{we get } a_j = e_j$$

$$\text{var}(y_j) = \lambda_j \quad \Sigma e_j = \lambda_j e_j$$

$$y_j = e_j^T X = \underbrace{e_{j1} X_1 + e_{j2} X_2 + \dots + e_{jp} X_p}_{\text{loadings on } j^{\text{th}} \text{ principal component}}$$

$\uparrow$   
 $j^{\text{th}}$  PC

$$\begin{aligned} \text{Total variance of population} &= \text{trace}(\Sigma) = \text{trace}(P \Lambda P^T) \\ &= \text{tr}(\Lambda P^T P) \\ &= \text{tr}(\Lambda) \\ &= \sum_{j=1}^p \lambda_j \end{aligned}$$

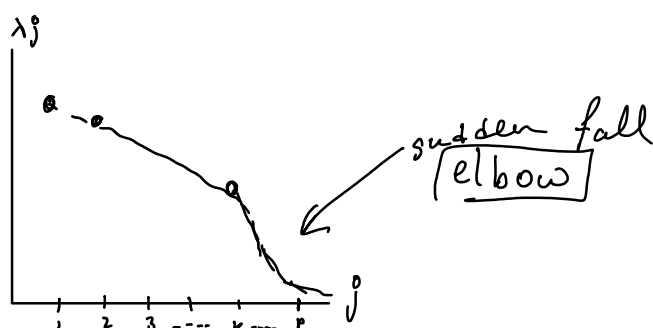
The  $j^{\text{th}}$  PC captures  $\lambda_j$  of the total var

$$\frac{\lambda_j}{\sum_{j=1}^p \lambda_j}$$

from the first  $k$  PC the var captured is  $\frac{\lambda_1 + \dots + \lambda_k}{\sum_{j=1}^p \lambda_j} \times 100\%$ .

$$\left[ \text{cov}(y_j, y_k) = 0 \quad \therefore \text{independent up to linearity} \right] \text{ hence this}$$

we find the smallest integers  $k$ ,  $1 \leq k \leq p$  at which this % is large enough to almost capture population total var



PCA is able to capture only linear variations when  $|E| \sim 0$ , PCA is not good here

What to do? Kernel trick

Ex  $\Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$  is given ( $p=3$ )

$$\lambda_1 = 5.83$$

$$\lambda_2 = 2$$

$$\lambda_3 = 0.17$$

$$\frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \times 100 \sim 98\%$$

$\therefore \lambda_1, \lambda_2$  are PC

$$e_1 = \begin{pmatrix} 0.383 \\ -0.929 \\ 0 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$e_3 = \begin{pmatrix} 0.929 \\ 0.383 \\ 0 \end{pmatrix}$$

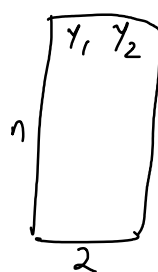
$$y_1 = e_1^T X = 0.383X_1 - 0.924X_2$$

$$y_2 = e_2^T X = X_3$$

sample data matrix



PC data matrix



Practically

We will have a data  $n \times p$  given to us  
we have sample var cov matrix  $(S_{p \times p})$

Asymptotic theory  
of PCA (not in course)

$$\hat{\lambda}_j \quad \hat{e}_j$$

$$(S \hat{e}_j = \hat{\lambda}_j \hat{e}_j)$$

$$\hat{y}_j = \hat{e}_j^T X$$

Suppose population is  $N_p(\mu, \Sigma)$

$$\sqrt{n}(\hat{\lambda}_j - \lambda_j) \xrightarrow[n \rightarrow \infty]{p \Lambda p^T} N(0, 2\lambda_j^2)$$

$$\left| \frac{\hat{\lambda}_j - \lambda_j}{\sqrt{\frac{2\lambda_j^2}{n}}} \right| \geq z_{\alpha/2}$$

reject  $H_0: \lambda_j = \hat{\lambda}_j$  at  
 $\alpha$ -level of significance  
 $H_1: \lambda_j \neq \hat{\lambda}_j$

CI for  $\lambda_j$

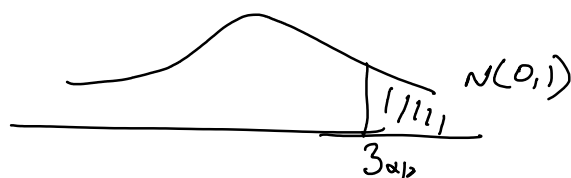
$$\left| \frac{\hat{\lambda}_j - \lambda_j}{\sqrt{\frac{2}{n}} \lambda_j} \right| \leq z_{\alpha/2}$$

$$-z_{\frac{\alpha}{2}} \sqrt{\frac{2}{n}} \leq \frac{\hat{\lambda}_j - \lambda_j}{\lambda_j} \leq \sqrt{\frac{2}{n}} z_{\alpha/2}$$

$$\left(1 - z_{\frac{\alpha}{2}} \sqrt{\frac{2}{n}}\right) \lambda_j \leq \hat{\lambda}_j \leq \left(1 + \sqrt{\frac{2}{n}} z_{\alpha/2}\right) \lambda_j$$

$$\lambda_j \leq \frac{\hat{\lambda}_j}{1 - z_{\frac{\alpha}{2}} \sqrt{\frac{2}{n}}}, \quad \lambda_j \geq \frac{\hat{\lambda}_j}{1 + \sqrt{\frac{2}{n}} z_{\alpha/2}}$$

$$\lambda_j \xrightarrow[n \rightarrow \infty]{CI} \left[ \frac{\hat{\lambda}_j}{1 + \sqrt{\frac{2}{n}} z_{\alpha/2}}, \frac{\hat{\lambda}_j}{1 - z_{\alpha/2} \sqrt{\frac{2}{n}}} \right]$$



as  $n \rightarrow \infty$   $\rightarrow p \times 1$

$$\sqrt{n}(\hat{e}_j - e_j) \xrightarrow[n \rightarrow \infty]{} N_p(0, E_j)$$

$$E_k = \sum_{\substack{j=1 \\ j \neq k}}^p \frac{\lambda_k}{(\lambda_j - \lambda_k)^2} e_j e_j^T$$

If there is a dimension / units attached to the data points,  
like km, sec etc then standardise each feature col

and do PCA in that matrix

$$\frac{x_i - \bar{x}_i}{\sqrt{\sigma_i^2}}$$